Exercise 8

Haoran Xu

```r
library(dplyr) # A Grammar of Data Manipulation
library(ggplot2) # Create Elegant Data Visualisations Using the Grammar of Graphics
library(kableExtra) # Construct Complex Table with 'kable' and Pipe Syntax
library(mlogit) # Multinomial Logit Models
```

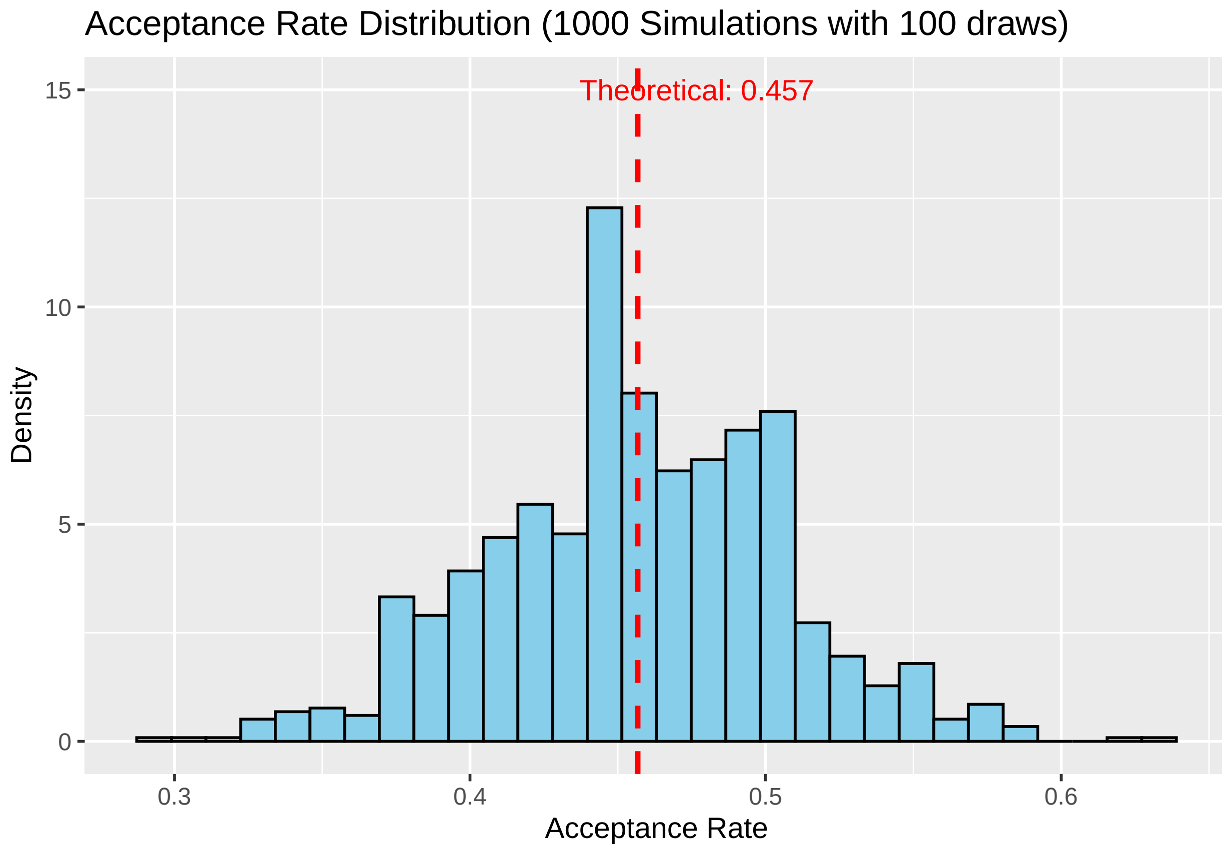
```r
data("Heating")
```

```r
H <- mlogit.data(Heating,
                 shape = "wide",
                 choice = "depvar",
                 varying = c(3:12))
```

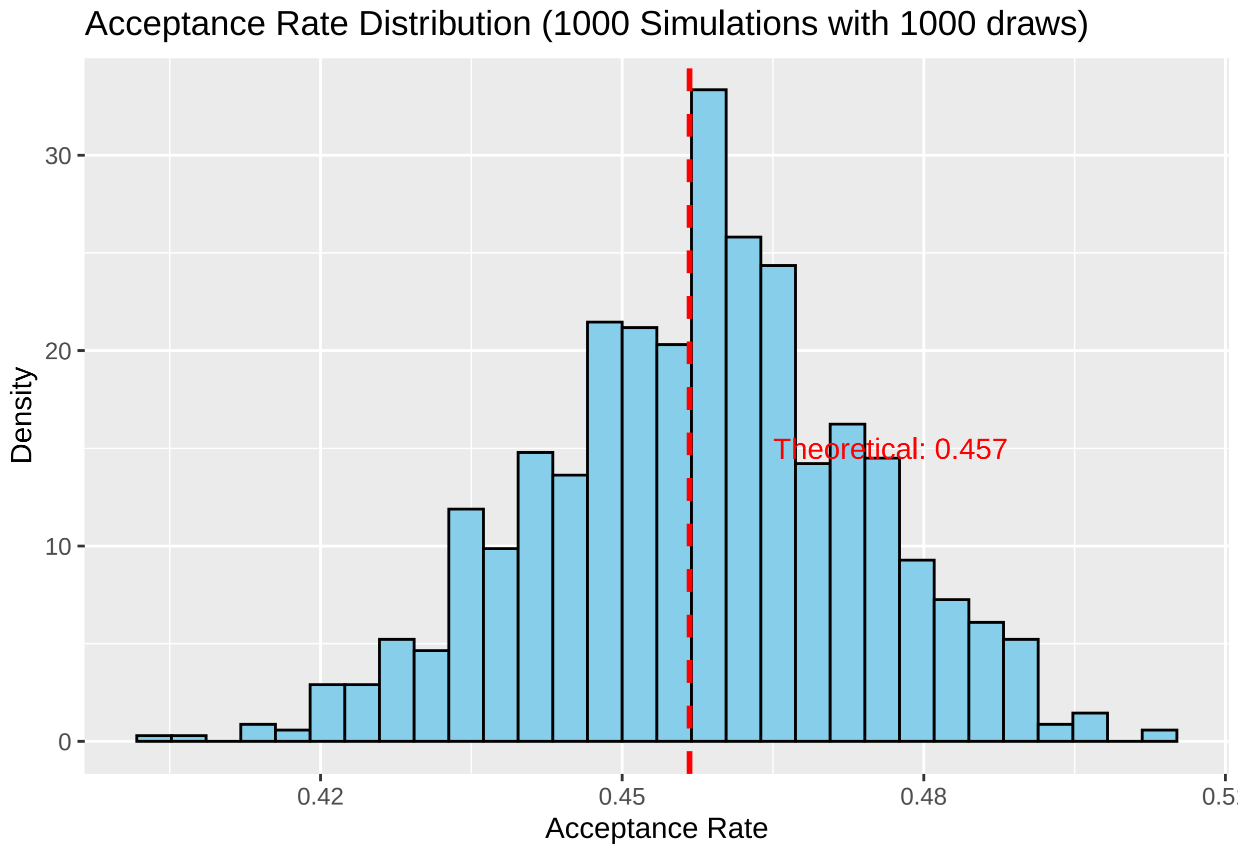
1. Repeat the illustrative accept-reject simulation multiple times (e.g., `s` = 1000) using `r` = 100 and summarize the results in the form of a histogram or frequency polygon. Next, do the same experiment butwith `r` = 1000 and `r` = 5000. Compare to the exact area and discuss your results. Hint: You can simultaneously draw `s` × `r` random numbers and then collect them in `s` groups (see package {dplyr} functions `group_by()` and `summarize()`).

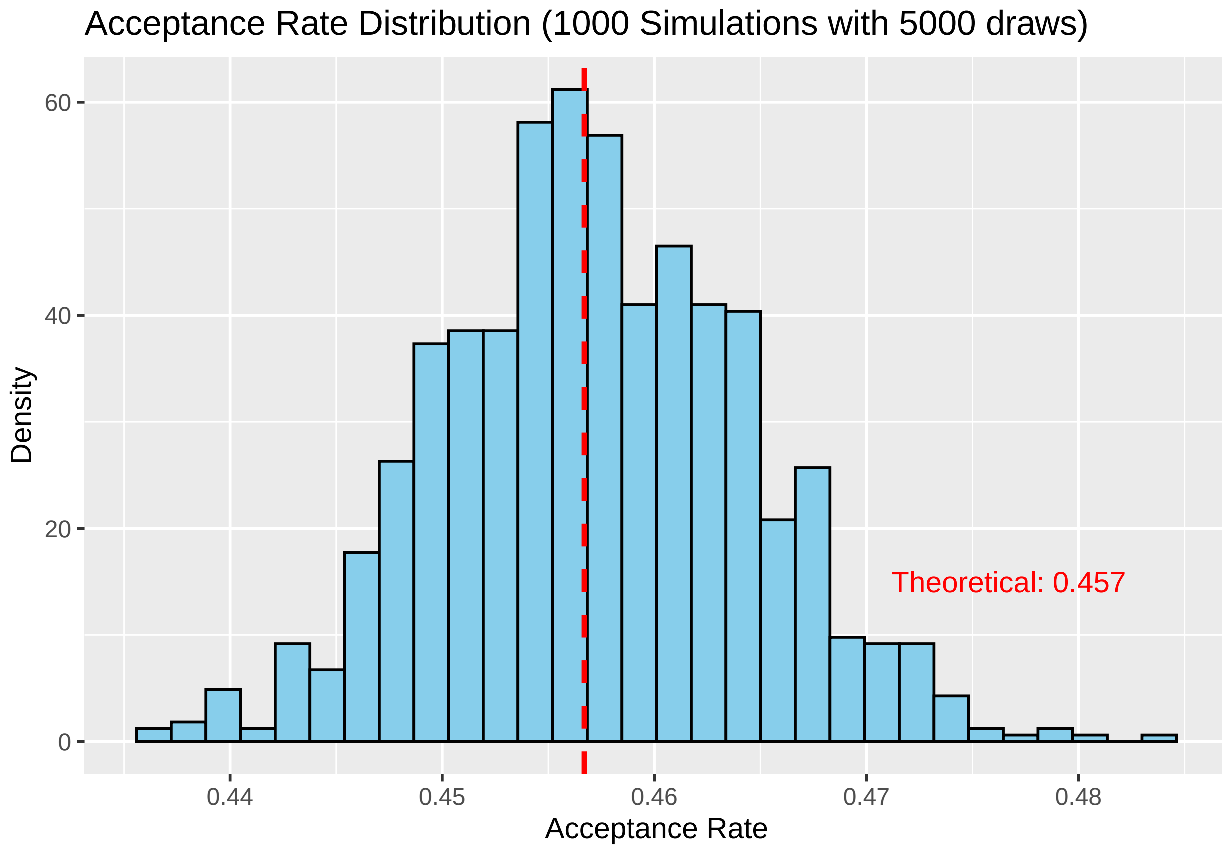
```r
# Set seed for replicability (this defines a starting number from which the sequence of random numbers
set.seed(35739)

norm_pdf <- data.frame(x = seq(0.2,
                               1.2,
                               0.01)) %>%
  mutate(f = dnorm(x,
                   mean = 0,
                   sd = 0.5))

# setting parameters
r <- 100
s <- 1000   # Number of simulations
xmin <- 0.2
xmax <- 1.2
fmax <- max(norm_pdf$f)  # Max height of PDF

sim_pdf <- data.frame(
  sim_group = rep(1:s, each = r),  # Create 1000 groups of 100 draws
  x = runif(s * r, xmin, xmax),     # All x-values
  fsim = runif(s * r, 0, fmax)      # All y-values
)

sim_pdf <- sim_pdf %>%
```

```r
  mutate(
    f = dnorm(x, mean = 0, sd = 0.5),
    status = ifelse(fsim <= f, "accept", "reject")
  )

# Calculate acceptance rate per simulation group
accept_rates <- sim_pdf %>%
  group_by(sim_group) %>%
  summarize(accept_rate = sum(status == "accept") / r)

# Theoretical acceptance probability (for comparison)
theoretical_prob <- integrate(
  function(x) dnorm(x, 0, 0.5),
  lower = xmin,
  upper = xmax
)$value / ((xmax - xmin) * fmax)

# Plot histogram of acceptance rates
ggplot(accept_rates, aes(x = accept_rate)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 30,
    fill = "skyblue",
    color = "black"
  ) +
  geom_vline(
    xintercept = theoretical_prob,
    color = "red",
    linetype = "dashed",
    linewidth = 1
  ) +
  labs(
    title = "Acceptance Rate Distribution (1000 Simulations with 100 draws)",
    x = "Acceptance Rate",
    y = "Density"
  ) +
  annotate(
    "text",
    x = theoretical_prob + 0.02,
    y = 15,
    label = paste0("Theoretical: ", round(theoretical_prob, 3)),
    color = "red"
  )
```



The codes for $r = 1000$ and $r = 5000$ are similar, here the histograms for each are listed.


Acceptance Rate Distribution (1000 Simulations with 1000 draws)
Density
30
20
10
0
Theoretical: 0.457
0.42
0.45
0.48
0.51
Acceptance Rate



The simulations demonstrate that increasing the number of draws per trial (r) significantly reduces variability in acceptance rate estimates. For $r = 100$, the histogram of acceptance rates across $s = 1000$ trials shows wide dispersion around the theoretical value (the exact area under the target density), reflecting high sampling variability. With $r = 1000$, the distribution narrows sharply, clustering tightly around the theoretical rate, and for $r = 5000$, the histogram becomes nearly unimodal and symmetric, closely aligning with the exact area. This illustrates the **Law of Large Numbers**: larger r yields more precise approximations of the true probability, while smaller r introduces greater stochastic noise. The results validate the accept-reject method's consistency, as empirical estimates converge to the theoretical value with increased sampling effort.

2. Re-estimate the multinomial probit model in this chapter, still with `r` = 50 but using a different random seed. Do this several times. The estimated values of the parameters will change, which is to be expected. . . but which ones change more? Why do you think this happens?

3. Use the models you estimated in Question 2 and repeat the scenario where the installation cost of gas central systems increases by 15%.What can you say about the patterns of substitution when you calculate them with different models? What would you do to be more confident about the estimated patterns of substitution?

4. Re-estimate the multinomial probit model in this chapter with `r` = 50 and the seed set to 3245, but change the reference level to "ec". What happens? What do you think are the implications for experimenting with multinomial probit model specifications?